

Data Engineering Assignment (Scala + Spark)

Domain: Transport Analytics

Dataset Schema

trip_data.csv: trip_id (Int), vehicle_id (String), driver_id (String), route_id (String), city (String), distance_km (Double), fare_amount (Double), trip_date (Date), start_time (String), end_time (String)

vehicle_data.json: vehicle_id (String), vehicle_type (String), capacity (Int), fuel_type (String)

Section 1: Scala Basics

- Explain val vs var, case classes, and higher-order functions.
- Create a case class Trip.
- Filter trips with fare > 150.
- Perform list operations: sum, average, max.

Section 2: Spark Basics

- Create SparkSession.
- Read CSV and JSON datasets.
- Perform select, filter, and create derived columns.

Section 3: Transformations & Joins

- Join trip and vehicle datasets.
- Create fare_per_km column.

Section 4: Aggregations

- Total trips per city.
- Average fare per city.
- Maximum distance per driver.
- Aggregations per vehicle type.

Section 5: Window Functions

- Top 2 highest fare trips per city.

- Running total of fare per driver.
- Previous trip fare using lag.

Section 6: Business Scenarios

- Find most profitable route.
- Identify drivers with frequent short trips.
- Find peak hour for trips.

Section 7: Bonus

- Write data partitioned by city in parquet format.
- Explain caching and broadcast joins.